# MATH 345 Skeleton Notes
# Unit 4: Orthogonality and least squares
# Section 4.3: Least squares as projection

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Revisiting a requested output as equations (U2-LO1, U2-LO3) asked whether a target vector $\mathbf{b}$ is reachable as $A\mathbf{x}$.
Recall the matrix and target from that activity:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}.$$

For

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix},$$

we have

$$A\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_1 + x_2 \end{bmatrix}.$$

Every reachable output therefore has third coordinate equal to the sum of its first two coordinates. The target $\mathbf{b}$ is not reachable because $4 \neq 1 + 2$.
More generally, when $\mathbf{b} \notin \operatorname{col}(A)$, no input reaches the target exactly. Least squares replaces the unreachable target by the closest reachable output.

### Definition: Least-squares solution, fitted vector, and residual

Let $A$ be an $m \times n$ matrix and let $\mathbf{b} \in \mathbb{R}^m$.

A [ ] is a vector $\widehat{\mathbf{x}} \in \mathbb{R}^n$ that minimizes

$$\|A\mathbf{x} - \mathbf{b}\|^2$$

over all $\mathbf{x} \in \mathbb{R}^n$.
The vector

$$\widehat{\mathbf{b}} = A\widehat{\mathbf{x}}$$

is the [ ]. The vector

$$\mathbf{r} = \mathbf{b} - \widehat{\mathbf{b}} = \mathbf{b} - A\widehat{\mathbf{x}}$$

is the [ ].
The quantity

$$\|A\mathbf{x} - \mathbf{b}\|^2 = \sum_{i=1}^{m} \big((A\mathbf{x})_i - b_i\big)^2$$

is the sum of the squares of the coordinate-wise errors. This explains the name least squares. Minimizing the norm or its square gives the same minimizers.
If $\mathbf{b} \in \mathrm{col}(A)$, the minimum error is 0, and least squares agrees with solving $A\mathbf{x} = \mathbf{b}$ exactly. If the target is not reachable, the fitted vector $\widehat{\mathbf{b}}$ is the closest reachable output.

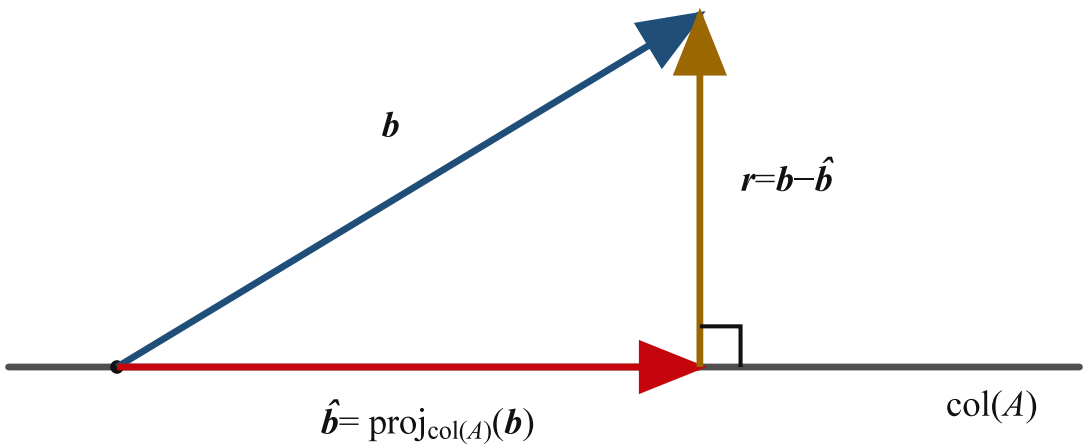



**Figure.** The target vector $\mathbf{b}$, its fitted vector $\widehat{\mathbf{b}}$ in $\mathrm{col}(A)$, and the perpendicular residual $\mathbf{r}$.

The residual is

$$\mathbf{r} = \mathbf{b} - \widehat{\mathbf{b}}.$$

Projection theorem guarantees a closest vector $\widehat{\mathbf{b}} \in \mathrm{col}(A)$. Since $\widehat{\mathbf{b}}$ lies in the column space, at least one coefficient vector $\widehat{\mathbf{x}}$ satisfies

$$A\widehat{\mathbf{x}} = \widehat{\mathbf{b}}.$$

Therefore the least-squares fitted vector is

$$\widehat{\mathbf{b}} = A\widehat{\mathbf{x}} = \mathrm{proj}_{\mathrm{col}(A)}(\mathbf{b}).$$

Its residual

$$\mathbf{r} = \mathbf{b} - A\widehat{\mathbf{x}}$$

is orthogonal to $\mathrm{col}(A)$.

**Fact: $A^T$ checks column orthogonality**

Write

$$A = \begin{bmatrix} \mathbf{a}_1 & \cdots & \mathbf{a}_n \end{bmatrix}.$$

For any vector $\mathbf{r} \in \mathbb{R}^m$,

$$A^T\mathbf{r} = \begin{bmatrix} \mathbf{a}_1^T\mathbf{r} \\ \vdots \\ \mathbf{a}_n^T\mathbf{r} \end{bmatrix}.$$

Therefore

$$A^T\mathbf{r} = \mathbf{0}$$

exactly when $\mathbf{r}$ is orthogonal to every column of $A$. Equivalently,

$$\mathrm{col}(A)^\perp = \mathrm{null}(A^T).$$

**Theorem: Least squares and the normal equations**

A vector $\widehat{\mathbf{x}}$ is a least-squares solution exactly when the residual

$$\mathbf{r} = \mathbf{b} - A\widehat{\mathbf{x}}$$

is orthogonal to $\mathrm{col}(A)$. Equivalently,

$$A^T\left(\mathbf{b} - A\widehat{\mathbf{x}}\right) = \mathbf{0}.$$

Thus the least-squares solutions are exactly the solutions of

the

$$A^TA\widehat{\mathbf{x}} = A^T\mathbf{b}.$$

**Why is this true?**

Every candidate fitted vector $A\mathbf{x}$ lies in $\mathrm{col}(A)$. By Projection theorem, $A\widehat{\mathbf{x}}$ is closest to $\mathbf{b}$ exactly when

$$\mathbf{b} - A\widehat{\mathbf{x}} \in \mathrm{col}(A)^\perp.$$

By $A^T$ checks column orthogonality, this is equivalent to

$$A^T\left(\mathbf{b} - A\widehat{\mathbf{x}}\right) = \mathbf{0}.$$

Expanding and rearranging gives

$$A^T\mathbf{b} - A^TA\widehat{\mathbf{x}} = \mathbf{0},$$

or

$$A^TA\widehat{\mathbf{x}} = A^T\mathbf{b}.$$

**Warning: Unique fitted vector, possibly nonunique coefficients**

The fitted vector $\widehat{\mathbf{b}}$ is unique because orthogonal projection onto a subspace is unique. The coefficient vector $\widehat{\mathbf{x}}$ need not be unique. For any $\mathbf{z} \in \text{null}(A)$,

$$A\left(\widehat{\mathbf{x}} + \mathbf{z}\right) = A\widehat{\mathbf{x}}.$$

When the columns of $A$ are linearly independent, $\text{null}(A) = \{\mathbf{0}\}$ by the matrix characterization of linear independence, so the least-squares coefficient vector is unique in that case.

**Fact: Independent columns and $A^TA$**

Let $A$ be an $m \times n$ matrix. Then $A^TA$ is invertible if and only if the columns of $A$ are linearly independent.

**Why is this true?**

If

$$A^TA\mathbf{x} = \mathbf{0},$$

then

$$0 = \mathbf{x}^TA^TA\mathbf{x} = \|A\mathbf{x}\|^2.$$

Therefore $A\mathbf{x} = \mathbf{0}$. If the columns of $A$ are linearly independent, then

$$\text{null}(A) = \{\mathbf{0}\},$$

so $\mathbf{x} = \mathbf{0}$. Thus $\text{null}(A^T A) = \{\mathbf{0}\}$, and the square matrix $A^T A$ is invertible. Conversely, suppose $A^T A$ is invertible and

$$A\mathbf{x} = \mathbf{0}.$$

Multiplying by $A^T$ gives

$$A^T A\mathbf{x} = \mathbf{0}.$$

Since $A^T A$ is invertible, $\mathbf{x} = \mathbf{0}$. Therefore

$$\text{null}(A) = \{\mathbf{0}\},$$

so the columns of $A$ are linearly independent.

**Example: Unit 2 target revisited (U4-LO4, U4-LO5)**
Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}.$$

1. Recall why $A\mathbf{x} = \mathbf{b}$ has no exact solution.
2. Form $A^T A$ and $A^T \mathbf{b}$.
3. Solve the normal equations for $\widehat{\mathbf{x}}$.
4. Compute the fitted vector

$$\widehat{\mathbf{b}} = A\widehat{\mathbf{x}}$$

and the residual

$$\mathbf{r} = \mathbf{b} - \widehat{\mathbf{b}}.$$

5. Check that

$$A^T\mathbf{r} = \mathbf{0}.$$

6. Compute $\|\mathbf{r}\|^2$.

7. Which vector is the closest reachable output: $\hat{\mathbf{x}}$ or $\widehat{\mathbf{b}}$?

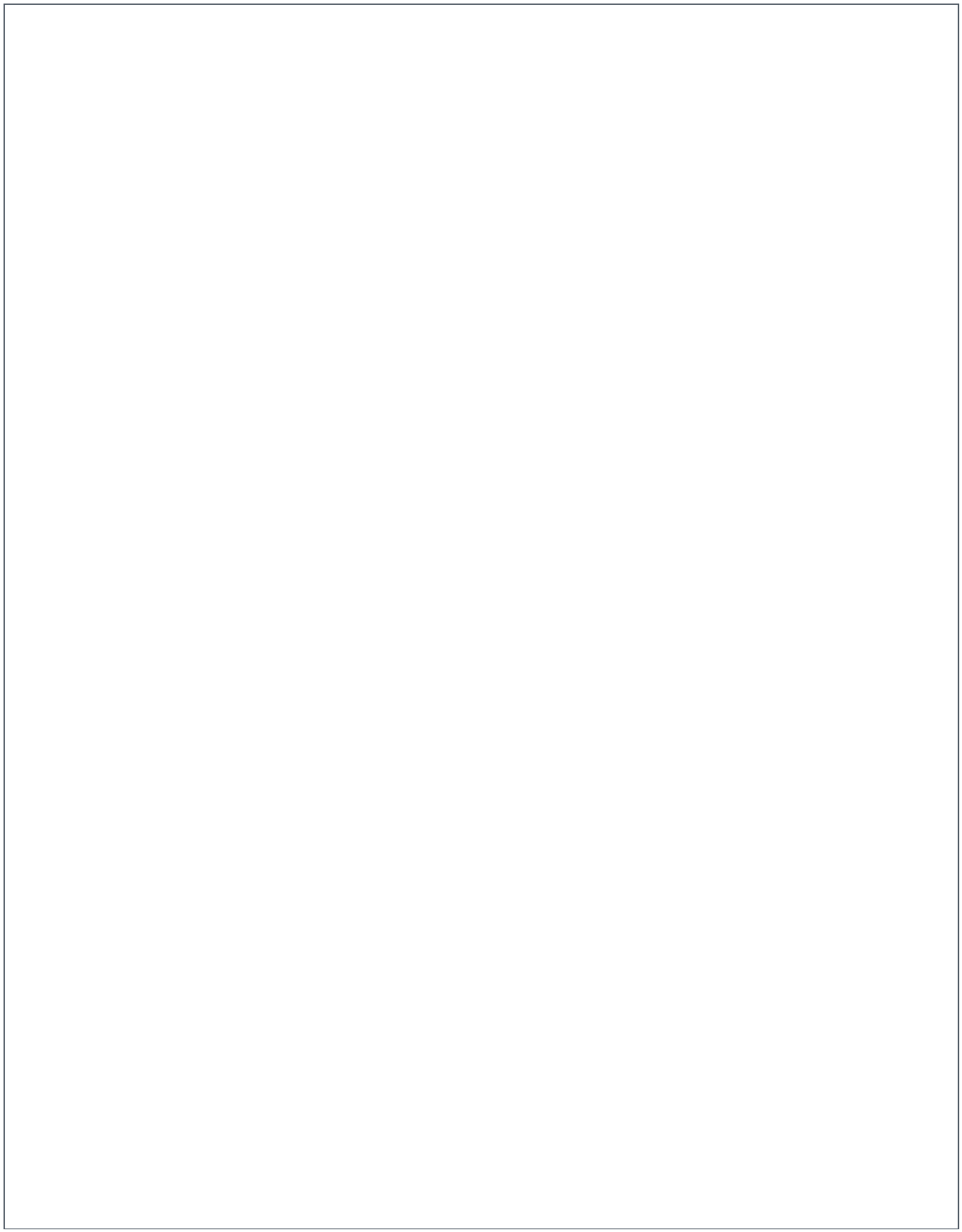
8. The residual is not zero. Explain why this does not mean least squares failed.

**Activity: Least squares and residual orthogonality (U4-LO4, U4-LO5)**

Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 2 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

1. Find the least-squares solution $\hat{\mathbf{x}}$ by solving the normal equations.
2. Compute the residual $\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}}$ and check that it is orthogonal to every column of $A$.



**Activity: Least squares with three unknowns (U4-LO4, U4-LO5)**

Let

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}.$$

Find the least-squares solution $\widehat{\mathbf{x}}$.

**Linear regression**

An important practical application of least squares is

[ ], which fits a linear model to observed data to describe relationships and make predictions. Suppose data are given as pairs

$$(t_1, y_1), \ldots, (t_m, y_m).$$

A line model predicts

$$\hat{y} = c_0 + c_1 t.$$

The coefficient $c_0$ is the [ ], and $c_1$ is the [ ].

At input $t_i$, the predicted value is

$$\hat{y}_i = c_0 + c_1 t_i.$$

The [ ] for the $i$-th data point is

$$r_i = y_i - \hat{y}_i.$$

Linear regression chooses $c_0$ and $c_1$ to make

$$r_1^2 + \cdots + r_m^2$$

as small as possible.
Linear regression is the same closest-reachable-output problem as Unit 2 target revisited (U4-LO4, U4-LO5), but the columns of the matrix now represent model features.

**Example: A line model as a matrix product (U4-LO4)**
Consider the data points

$$(0, 1), \qquad (1, 2), \qquad (2, 2),$$

and the line model

$$\hat{y} = c_0 + c_1 t.$$

1. Write the three predicted values in terms of $c_0$ and $c_1$.
2. Define

$$\mathbf{c} = \begin{bmatrix} c_0 \\ c_1 \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

   Build a matrix $X$ such that $X\mathbf{c}$ is the vector of predicted values.
3. What does the first column of $X$ represent?

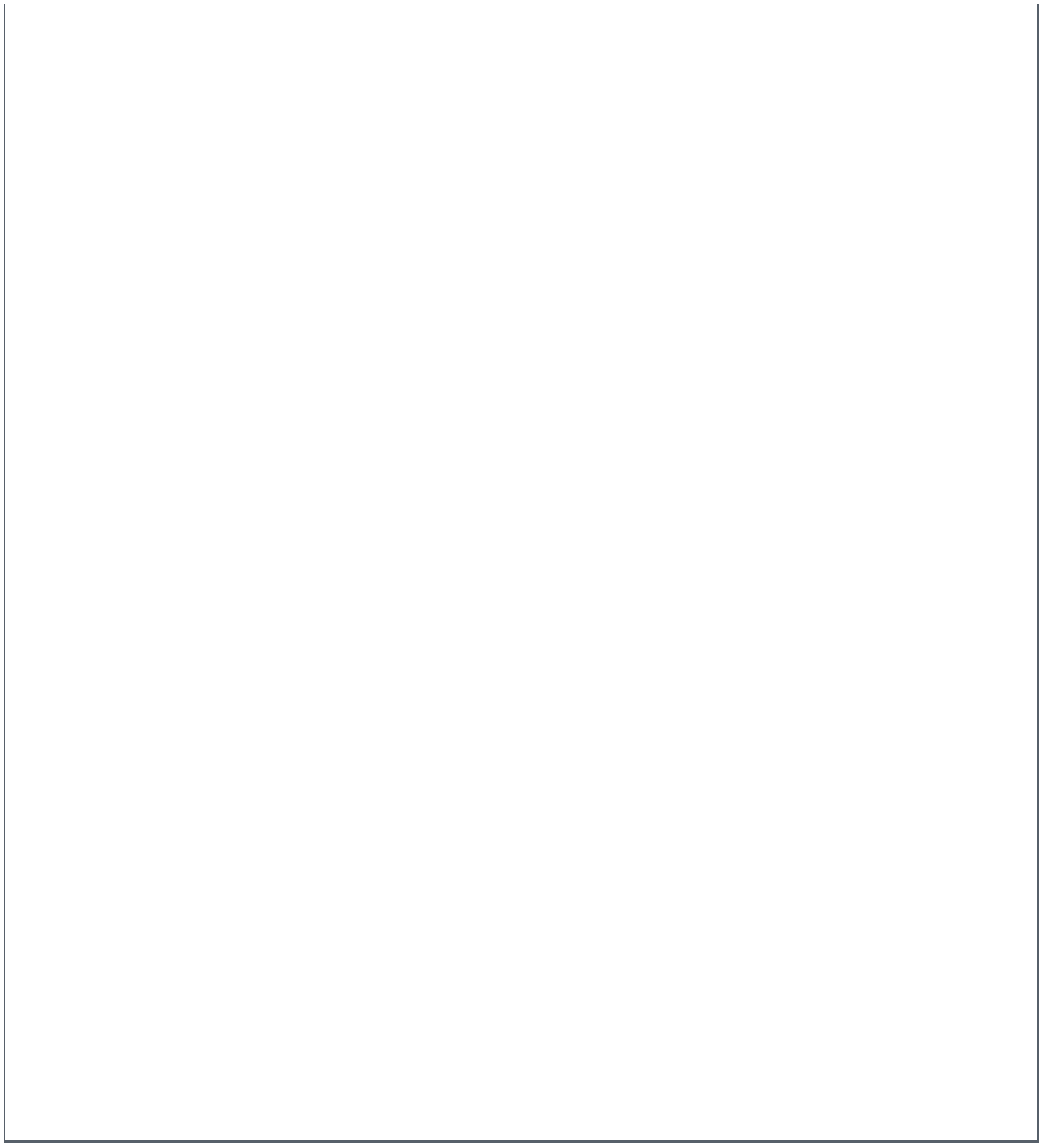
4. What does the second column of $X$ represent?
5. Write the residual vector

$$\mathbf{r} = \mathbf{y} - X\mathbf{c}.$$

6. Explain why

$$\|\mathbf{r}\|^2 = \sum_{i=1}^{3} (y_i - \hat{y}_i)^2$$

is the sum of the squared vertical errors in the data plot.



See the figure for the data points, fitted line, and vertical residuals.
For general data $(t_i, y_i)$, define the design matrix

$$X = \begin{bmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix}, \qquad \mathbf{c} = \begin{bmatrix} c_0 \\ c_1 \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}.$$

Then $X\mathbf{c}$ is the vector of predicted values. Linear regression is the least-squares problem

$$\min_{\mathbf{c}} \|X\mathbf{c} - \mathbf{y}\|^2.$$

The matrix $X$ is called the [ ]. Each column represents one feature used by the model.

**Warning: Two different geometric pictures**

The projection occurs in $\mathbb{R}^m$:

$$\hat{\mathbf{y}} = X\hat{\mathbf{c}} = \text{proj}_{\text{col}(X)}(\mathbf{y}).$$

In the usual $t$-$y$ data plot, the residuals are vertical differences

$$r_i = y_i - \hat{y}_i.$$

They are not generally perpendicular distances from the plotted points to the fitted line.

**Activity: Fit a line to three data points (U4-LO4, U4-LO5)**

Continue with the design matrix and data vector from A line model as a matrix product (U4-LO4). Recall that the data points are

$$(0, 1), \qquad (1, 2), \qquad (2, 2).$$

1. Use Least squares and the normal equations to form

$$X^T X\hat{\mathbf{c}} = X^T\mathbf{y}.$$

**Activity: Fit a line to three data points (U4-LO4, U4-LO5) (continued)**

2. Solve for

$$\hat{\mathbf{c}} = \begin{bmatrix} \hat{c}_0 \\ \hat{c}_1 \end{bmatrix}.$$

3. Write the fitted line.

4. Compute the fitted-value vector

$$\hat{\mathbf{y}} = X\hat{\mathbf{c}}$$

   and the residual vector

$$\mathbf{r} = \mathbf{y} - \hat{\mathbf{y}}.$$

5. Check that

$$X^T\mathbf{r} = \mathbf{0}.$$

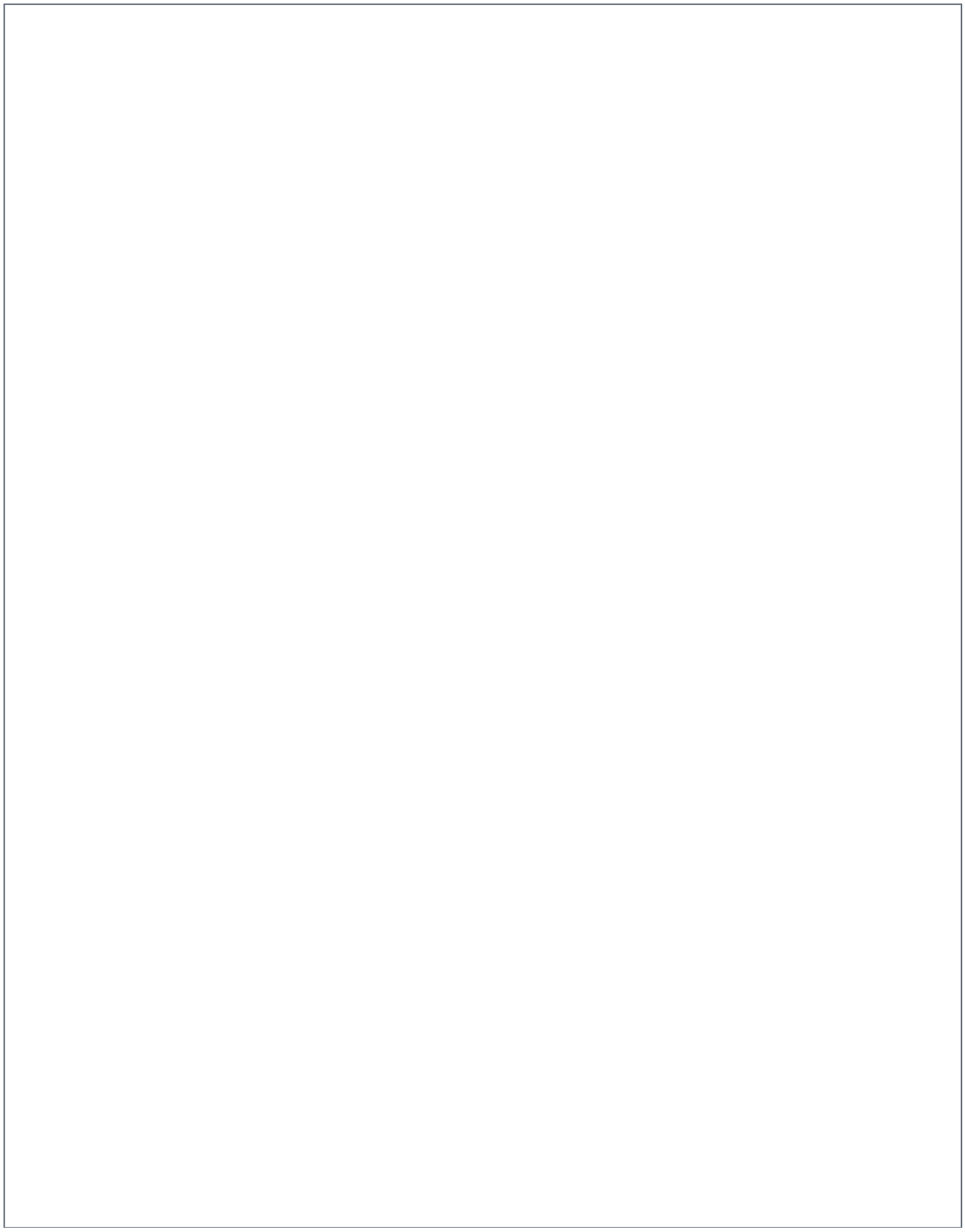
6. Write the two scalar equations contained in $X^T\mathbf{r} = \mathbf{0}$.

7. Compute

$$\|\mathbf{r}\|^2.$$

8. Which object is the fitted vector: $\hat{\mathbf{c}}$ or $\hat{\mathbf{y}}$?

9. In the $t$-$y$ plot, are the residual segments vertical or perpendicular to the fitted line?



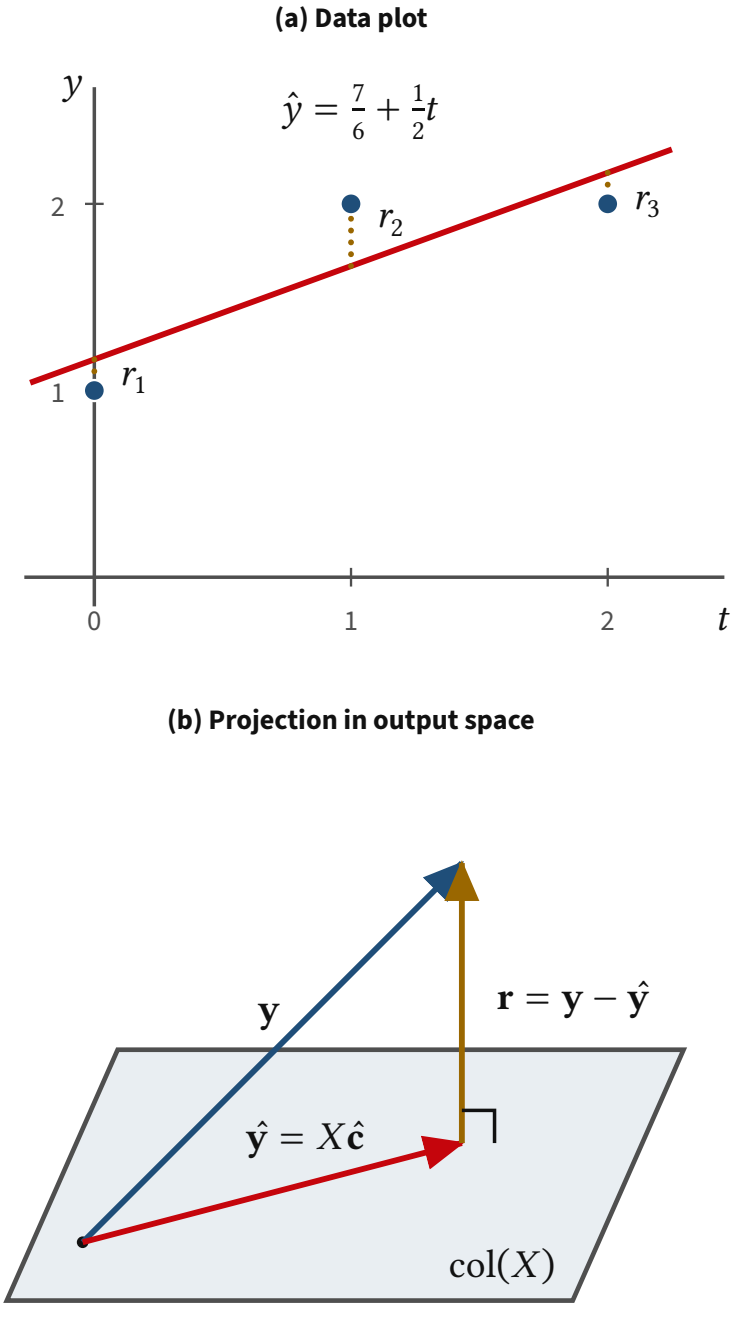



**Figure.** Two vertically arranged panels show the same linear regression. The upper panel (a) contains three data points, a fitted line, and vertical residual segments. The lower panel (b) shows the data vector y projected onto the column space of X, with a perpendicular residual r.

The two panels in the figure show different spaces: the upper panel (a) is the $t$-$y$ data plot, while the lower panel (b) is the output space $\mathbb{R}^3$.

**Activity: Reading regression code (U4-LO4, U4-LO7)**

The following code repeats the line fit from Fit a line to three data points (U4-LO4, U4-LO5). The expression `np.ones_like(t)` creates an array of ones with the same shape and data type as `t`. For the array `t` used below, it produces `array([1., 1., 1.])`. The call `np.column_stack([np.ones_like(t), t])` uses these two arrays as columns of the design matrix: the first column contains ones, and the second contains the input values in `t`. Thus

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

The NumPy function `np.linalg.lstsq(X, y, rcond=None)` returns a tuple containing the least-squares coefficient vector, an array of residual sums of squares, the rank of $X$, and its singular values (to which we will come back in Eigenvalues and second-order optimization). We use `[0]` to select the first item, the coefficient vector. The parameter `rcond` controls the relative numerical cutoff used to determine the matrix rank; `rcond=None` uses NumPy's default cutoff.

```
import numpy as np

t = np.array([0.0, 1.0, 2.0])
y = np.array([1.0, 2.0, 2.0])

X = np.column_stack([np.ones_like(t), t])

c = np.linalg.lstsq(X, y, rcond=None)[0]
yhat = X @ c
r = y - yhat
```

## Activity: Reading regression code (U4-LO4, U4-LO7) (continued)

```
X, c, yhat, r, X.T @ r, np.linalg.norm(r)**2
```

1. What is the shape of `X`?
2. Why does `c` have two entries?
3. What mathematical objects are stored in `yhat` and `r`?
4. Why is the residual check `X.T @ r` rather than `X @ r`?
5. What should `X.T @ r` be close to?
6. Does a nonzero residual vector mean the code failed?
7. What quantity is computed by `np.linalg.norm(r)**2`?

Explore least squares interactively with the Residual Squares mini-game.